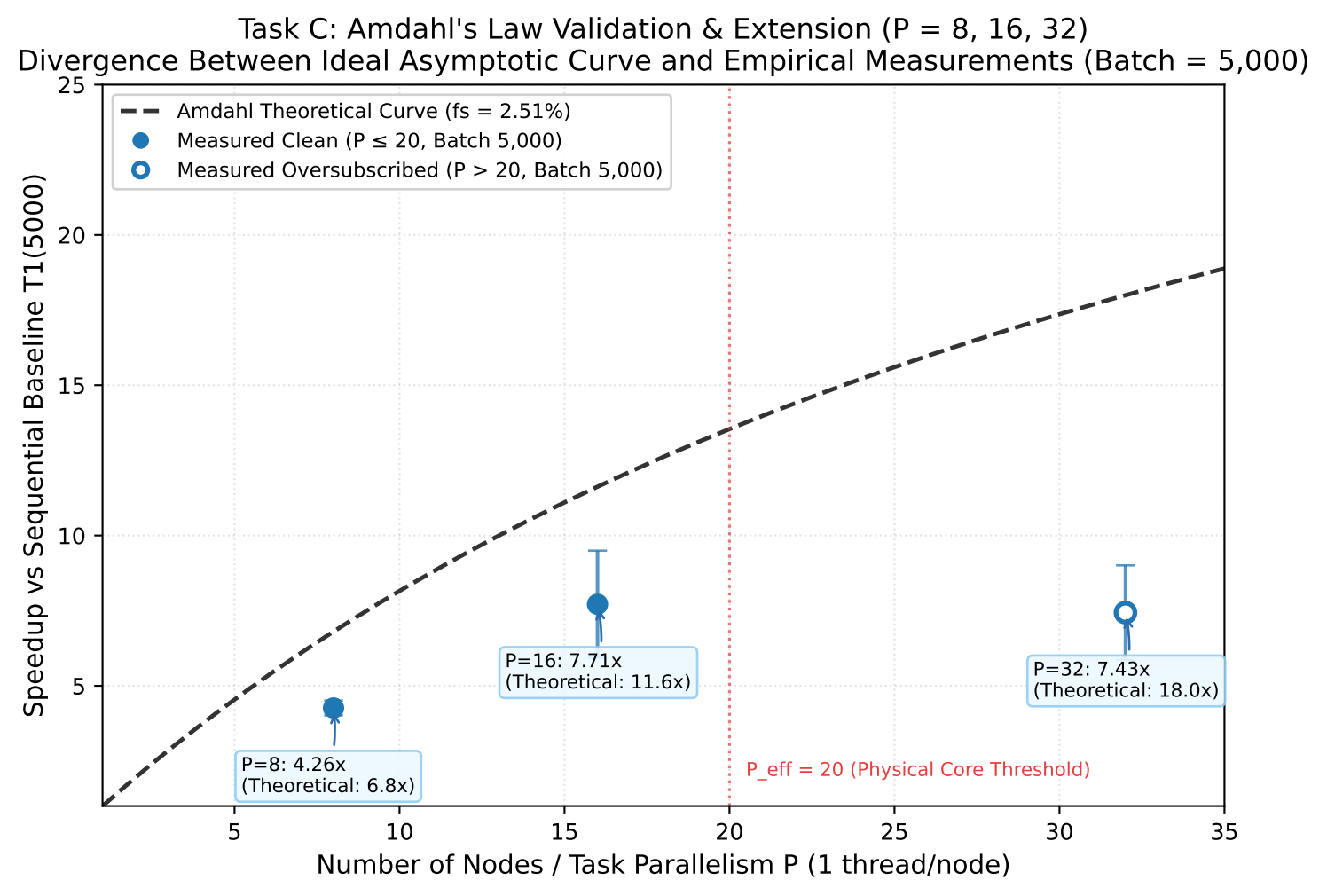
Task C: Amdahl's Law Validation & Extension (P = 8, 16, 32)
Divergence Between Ideal Asymptotic Curve and Empirical Measurements (Batch = 5,000)
Amdahl Theoretical Curve (fs = 2.51%)
Measured Clean (P ≤ 20, Batch 5,000)
Measured Oversubscribed (P > 20, Batch 5,000)
P=8: 4.26x
(Theoretical: 6.8x)
P=16: 7.71x
(Theoretical: 11.6x)
P=32: 7.43x
(Theoretical: 18.0x)
P_eff = 20 (Physical Core Threshold)
Speedup vs Sequential Baseline T1(5000)
Number of Nodes / Task Parallelism P (1 thread/node)
5
10
15
20
25
30
35